Saurabh Surashe

MTS III at Mavenir Systems

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Professional Summary

Embedded Software Engineer with 4+ years of experience developing production-grade Embedded Linux applications using Modern C++ and C for carrier-grade wireless communication systems. Experienced in embedded software development, protocol stack implementation, deterministic scheduling, low-level system programming, runtime optimization, and performance-critical software.

Interested in building embedded Linux platforms, firmware, networking software, and high-performance embedded systems, AI Infrastructure and C and C++ software development.

Core Competencies

Programming Languages: C++17, STL, C, Python, Java

Linux & Systems: Linux, POSIX, pthreads, Shared Memory IPC, Multithreading, GDB, Intel VTune

Networking & Protocols: TCP/IP, UDP, NB-IoT, MAC, RRC, S1AP, Wireshark, ASN.1

Software Engineering: Concurrent Programming, Distributed Systems, Performance Optimization, Low-Latency Systems, Carrier-Grade Software, Event-driven programming

AI / Machine Learning: Sentence Transformers, Semantic Search, Embeddings, TensorFlow, Keras

Development Tools: Git, Jenkins, Jira

Professional Experience

Mavenir Systems Pvt. Ltd. July 2022 – Present

Member of Technical Staff I → Member of Technical Staff III *Bengaluru, India*

- Develop embedded Linux software for carrier-grade real-time wireless communication systems using C++ and C, contributing across scheduler implementation, protocol stack development, performance optimization, debugging, and production software delivery.
- Redesigned latency-sensitive scheduler algorithms, improving uplink throughput by approximately **40%** while maintaining deterministic execution for real-time embedded workloads.
- Delivered more than **130 production software enhancements**, including embedded software features, protocol improvements, runtime optimizations, and production reliability fixes.
- Designed efficient **shared-memory IPC** mechanisms enabling low-latency communication between embedded software modules while improving runtime synchronization and scalability.
- Developed and maintained more than **60 production KPIs** providing runtime diagnostics, software observability, scheduler monitoring, and protocol performance analysis.
- Optimized CPU-intensive execution paths using Intel VTune, Linux perf, and GDB by identifying bottlenecks, reducing execution latency, and improving runtime efficiency.
- Performed root-cause analysis of complex production issues using runtime logs, protocol traces, memory analysis, Wireshark, and low-level debugging techniques.
- Collaborated with Layer-1, Platform, QA, DevOps, and customer engineering teams throughout software design, implementation, validation, deployment, and production support.
- Mentored junior engineers through technical discussions, debugging sessions, code reviews, and software design guidance.

Projects

AI-Powered Job Discovery & Resume Matching Platform | *Python, FastAPI, SQL, Sentence Transformers, SMTP*

- Developed a modular Python platform automating job discovery and semantic matching using the `all-MiniLM-L6-v2` transformer model to calculate resume-to-job description similarity scores.
- Built asynchronous scraping pipelines using `asyncio` to concurrently fetch and parse opportunities from major ATS engines including Workday, Greenhouse, Eightfold, and SmartRecruiters.
- Implemented highly optimized, in-memory semantic similarity search utilizing NumPy vector calculations, achieving sub-200ms scoring latency without database-side vector extensions.

- Designed clean REST APIs with FastAPI to support resume parsing, job indexing, user subscription profiles, and score-threshold configurations.
- Integrated an automated notification system sending formatted HTML job digests via SMTP for jobs matching custom score thresholds, deployed via serverless pipelines and background tasks.

Achievements

Mavenir Spotlight Award (Received Multiple Times) for delivering high-impact production features, technical excellence and successful customer software releases.

Co-authored: **"Privacy and Security Comparison of Web Browsers: A Review"**, AINA 2022 (*SpringerLink*)

Co-authored: **"Enhancing Cyber Security: Malicious web page detection using Machine and Deep Learning"**, AINA 2025 (*SpringerLink*)

Education

National Institute of Technology Surathkal — Karnataka
Master of Technology in Computational Data Science — *June, 2020 – May, 2022*

MIT College of Engineering Pune — Maharashtra
Bachelor of Engineering in Mechanical Engineering — *June, 2015 – May, 2019*